Next Generation Nutrition Apps: Utilising NHANES Data to provide an elevated user experience

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1. Introduction

Our goal was to create an interactive and personalized diet recommendation platform tailored to individual health profiles, nutritional needs, and lifestyles. Utilizing NHANES data, this system provides recommendations to mitigate health risks. We expanded upon existing methodology in disease prediction such as logistic regression, random forest and XGBoost.. Based on the literature we are focused on recommended variables such as BMI, daily sugar intake, demographic data etc. Our models were evaluated using accuracy measures which will be described in depth in the evaluation section, to ensure robustness. For tooling, we used python for the data cleaning, statistical analysis and machine learning modelling and a Flask based API to integrate with a web frontend. The web frontend is a single page application (SPA) built using Vue.js and will use D3 for visualizations. Our target audience includes health-conscious individuals, those with specific dietary needs, and clinicians.

By applying predictive modelling, our interactive tool will help users make necessary lifestyle and dietary changes. After providing the prediction criteria, i.e. the likelihood of disease, we will visualise this in an interactive slider or levers where the end user can change various features and visualise how this affects disease likelihood. While our final goal was to create an interactive, personalized nutritional recommendation and prediction platform prototype backed by clinical data, we built a prototype first in Microsoft PowerBI as the aspirational end state using synthetic prediction data and interactive visualisation, and began developing the webapp front end which is not yet dynamically updated, using static data for demonstrative purposes.

Our motivation stems from the urgent need to transform static health data into actionable insights that can prevent the onset of chronic illness. With the rise of diabetes and related conditions globally, it is no longer sufficient to describe risk; we must predict and prevent it. This work not only supports the shift toward proactive care but also addresses the equity gap by prioritizing accessibility, interpretability and clinical relevance.

2. Problem Definition

Current diet and nutrition apps like MyFitnessPal and Cronometer focus on food logging and calorie counting but often lack personalized, scientifically backed meal plans. According to Schoeppe et al. (2017), many apps miss incorporating behaviour change techniques and personalized feedback, limiting their effectiveness. Chen et al. (2017) emphasized that most apps provide generic advice, failing to tailor recommendations based on unique health data.

Our project aims to create a prototype for a personalised nutritional recommendation platform, based on clinical data, accessible to everyone. This will support health clinicians in providing evidence-based advice to at-risk patients who need lifestyle changes.

3. Literature Survey

Using NHANES data in personalized diet systems offers detailed dietary and health information, enabling algorithms to detect nutrient deficiencies and recommend tailored diets. For example, Fulgoni et al. (2019) created the Nestlé Nutrition Algorithm using NHANES data for personalized dietary advice. Guo (2024) and Liu (2024) highlighted vitamin D and B12 deficiencies as key disease indicators, stressing the need for a self-evaluation tool incorporating gender, age, race, weight, and smoking habits. Our approach will utilize decades of NHANES data for a more comprehensive analysis compared to prior studies. While literature links NHANES data to disease prediction and diet recommendations, combining predictive disease detection with personalized nutrition is less common. Amiri et al. (2023) demonstrated that machine learning models could predict optimal diets using NHANES data, showcasing AI's potential in personalizing nutrition advice.

Personalized nutrition interventions have shown positive health outcomes. A randomized controlled trial by Celis-Morales (2015) reported improved dietary habits and health indicators after 12 months. Success in this platform could reduce the risk of age-related and non-communicable diseases, potentially alleviating the burden on clinicians and hospitals.

From a front-end design perspective, Backonja et. al (2018) highlight the effectiveness of color coding, familiar formats and reference lines to enhance the data visualisation understanding and support personalized care. Zhang and Kaufman (2020) discuss that effective health data visualizations should adhere to design principles such as clarity and simplicity, appropriate use of color, i.e. green for low risk, red for high risk, amber for moderate risk. Zhang and Kaufman (2020) also recommend providing contextual information to help users interpret the data accurately, like explanation of risk levels and recommended actions. User engagement is crucial for personalized diet systems. Spring et al. (2013) found that personalized feedback and interactive features enhance user engagement. Mattila et al. (2010) proposed metrics like user retention, nutrient intake changes, and clinical health improvements as success measures. Our application will monitor these metrics to ensure user engagement.

Implementing personalized diet systems raises ethical considerations, particularly regarding data privacy and algorithmic bias. Mittelstadt et al. (2016) stress the importance of robust data protection protocols and algorithmic transparency. Compliance with global regulations like GDPR and the EU AI Act is essential. Despite the risks, the long-term benefits outweigh the initial costs. Ronto et al. (2017) suggest that improved public health outcomes and reduced healthcare expenditures due to better dietary habits make the investment worthwhile. Our project's minimal costs further enhance its feasibility.

3.i. Expected Innovations

Applications of AI in Nutrition Brown, L., & Davis, K. (2024): This review highlights several applications of artificial intelligence in the field of nutrition, including personalized nutrition and dietary assessment. The article details how AI can be incorporated into nutritional science, thereby informing the technological framework of your project. The review emphasizes the need for increased interdisciplinary collaboration, which your project can address by integrating expertise from both nutrition science and data analytics.

Machine Learning Model for Diet Recommendations Ahmed, S., & Ali, H. (2025): This study uses K-means clustering and Random Forest algorithms to analyse nutritional data and user profiles, providing personalized diet advice. Continuous updates of user data are essential to maintain recommendation accuracy.

Machine Learning Model for Diet Recommendation Awareness Zhu et al. (2018): Machine learning techniques offer advanced methods for analysing dietary patterns and predicting health outcomes. In this study, applied machine learning algorithms to NHANES dietary data to identify patterns associated with obesity, highlighting the potential of these approaches in developing personalized diet recommendations.

4. *Proposed Method*

4.1 Proposed Method - Intuition

What we see today in the health application ecosystem is a polarized availability of tools. On one end of the spectrum, commercialized platforms or subscription-based models lack the necessary clinical and prescriptive methodology and can be exclusive and not accessible to certain groups of people. Alternatively, the current platforms provided by government institutions lack an innovative approach to provide preventative care to the masses. (Centers for Disease Control and Prevention (n.d.)) Visualising NHANES data provided by the CDC is primarily a descriptive exercise, meaning it’s only demonstrating the data at face value, whereas we want to incorporate predictive modelling and recommendation modelling, identifying if patients may be at risk for certain diseases.

4.2 Proposed Method - Detailed Approach – Algorithmic, Backend & Frontend Integration

Our approach includes extensive data cleaning activities; predictive analytics surfaced via an API integration with a front-end user interface to visualize disease risk and dietary recommendations to ensure the risk is mitigated or lowered. NHANES data is free to download and is available to the public for use. NHANES datasets appear in two-year cycles, and we selected data from 1999-2023 and various attributes are captured in different datasets. For example, in the demographic's dataset, this is where attributes like age, gender, and ethnicity would be captured, but in laboratory data we would find biometric data such as glucose, triglycerides, insulin levels, etc. We found that not every column or attribute is accounted for across multiple years and requires a programmatic way to combine data sources from multiple years, accounting for attributes necessary for prediction and recommendation. We built a python script which identifies similar columns across the multiple cycles of NHANES and then be combined and joined by the unique identifier SEQN, i.e., an individual record.

We built upon Semerdjian & Frank (2017) research to identify essential datapoints for diabetes prediction. Key attributes include Age, Gender, Race/Ethnicity, BMI, Waist Circumference, Blood Pressure, Cholesterol levels, Triglycerides, Fasting Glucose, HbA1c, physical activity, smoking habits, alcohol use, dietary data, and diabetes status. Both quantitative and categorical data are utilized, such as Sex, Race & Ethnicity, and activity levels. For prediction models, we experimented with training and testing logistic regression, XGBoost, and Random Forest Generator, and ultimately elected the XGBoost model given the accuracy output, discussed in *section 5*. The output of this prediction model informs the visualisation of diabetes risk, meanwhile our diet recommendations are static data points with the intention to evolve into recommendation modelling.

The orchestration of the prediction model requires both back-end and front-end integration. The backend API uses Python 3.13 and Flask which creates the environment to run and execute the code. This backend API serves as a predictive service for identifying the likelihood of a diabetes diagnosis using a pre-trained XGBoost model. The Flask API exposes the following endpoints:

1. *'GET /features': returns the list of input features expected by the model*
2. *'POST /predict': accepts feature input as a JSON and returns a prediction result for diabetes*

This result is then surfaced in the front end; we use a single page application (SPA) using Vue.js and D3 for visualizations. The benefits of using Vue.js include building UI with reusable components, which could be beneficial for maintenance and scalability. Vue also supports dynamic interfaces and live updates and is lightweight and fast. We have elected D3 for the customization ability and interactivity and animation, as well as being compatible with Vue. Since our data is multidimensional, D3 can support complex data structures with ease. For ease of external testing, we packaged our code in Docker to ensure other users could easily run the API and front end without required access to private GitHub instances where our team primarily built the code. This will also be beneficial in the future if we were to scale the solution.

5. *Evaluation*

5.1 Evaluation - Questions our experiments are designed to answer:

1. Which models provide the highest accuracy, precision and recall for diabetes prediction on the NHANES dataset?
2. Do users understand the visual risk indicators and associated health metrics
3. How does visual complexity impact user experience and comprehension?
4. Does the presentation of recommendations improve the likelihood of users acknowledging or considering lifestyle changes?

5.2a Evaluation - Algorithmic Experiments & Results:

For experimentation, as mentioned above we experimented with logistic regression, random forest classification, and XGBoost models for diabetes prediction, first focused on 2017-2018 NHANES data. In the future we hope to expand this platform to additional disease prediction and mitigation, like cardiovascular disease, lung disease or other common occurring disease. We encoded the target variable

(*Doctor_Diagnosed_Diabetes*) and input features from NHANES datasets. Before training logistic regression models, we had to address missing values and split the data into train and test sets. Systolic BP, Diastolic BP, and Avg Drinks Per Day had the highest percentage of missing values, ranging from 12%-16%. We imputed these missing values using the mean after splitting the data.

The logistic regression model has a weighted average 85% precision and 88% recall, which reflects the dominance of non-diabetics in the dataset, thus larger classes have more impact on the average. This 88% accuracy is misleading since this test dataset is highly imbalanced, most cases are non-diabetic, which is easy for the model to get those right, whereas the model ignores small classes like borderline diabetic. Thus, we tried different models what handle imbalance better, like XGboost and random forest classifier.

The overall accuracy of Random Forest Classification is 88%, but the model performs poorly on Borderline/Pre-Diabetic patients, likely due to the fact there is only 3.1% of those cases in the test dataset. It's reasonably good at identifying Diabetics and very strong for non-diabetic patients, which is dominant in the dataset (80% of the test dataset). *(Appendix: Table 1)*

The XGBoost model performs well in identifying non-diabetic cases (95.5% recall) and moderately in detecting diabetic cases (65% recall). However, it fails to identify borderline/prediabetic cases, with zero correct predictions. This gap suggests the need for class balancing techniques like SMOTE or class weighting. Upon selecting XGBoost Model for diabetes prediction we then expanded the dataset to 1999-2023 NHANES data. The data size is 113,249 rows and 25 columns. With this larger dataset, we first addressed the class imbalance of 'diabetes' versus 'no-diabetes'. To deal with the imbalance in the number of diabetic vs non-diabetic cases, we calculate the ratio of negative to positive samples in the training data. This ratio is then passed to the XGBoost classifier as *'scale_pos_weight'* to help balance the learning process and reduce bias toward the majority class. We separate the features (X) from the label (Y) and split the data into training and test sets, similar to the experiments above.

We also experimented with implementing a decision threshold to used to convert the model's predicted probability into a binary class (0 or 1), instead of the default 0.5, a higher threshold like 0.81 suggests the model is more conservative in predicting a 'positive' result (i.e. at risk of diabetes).

The accuracy results for XGBoost with extended data are as follows: When the model predicted diabetes, it was correct 68.93% of the time, in other words, ~31% of the *positive* predictions were false positives. The model correctly identified 71.26% of actual diabetes cases, meaning 29% of the diabetic individuals were missed, i.e. *false negatives*. Area under the Receiver Operating Characteristic (ROC) curve measures how the model distinguishes between classes across all possible thresholds. At 96.01%, the model is excellent at ranking positive examples higher than negative ones. This shows the model is well-calibrated and effective prior to thresholding.

What we observed is that logistic regression is sensitive to multicollinearity, which we anticipate in data such as health data and cannot handle missing values. Random forest models are black boxes and not as easily interpretable as logistic regression and slower and less efficient because they tend to use more memory and compute than XGBoost. In general, XGBoost models have higher predictive performance, handles feature interactions automatically and handles missing data with ease and can support imbalanced data.

Model	Precision	Recall	Accuracy
Logistic Regression (2017-2018 data)	85%	88%	**88%**
Random Forest (2017-2018 data)	81%	57%	**88%**

Table 1: Model Selection Comparison & Criteria

XGBoost (2017-2018 data)	72%	65%	**87.6%**
XGBoost (1999-2023 data)	68.93%	71.26%	**95.51%**

5.2b Evaluation - API Experiments & Results:

When integrating our model.json file with the API, we experienced issues with the model consistently returning 'false' for diabetes prediction, which was in large part due to the imbalance of data in NHANES dataset, thus motivating our rationale to perform the fine tuning activities noted above.

The model expects a full set of features, i.e. the features indicated in the XGBoost model, but the platform fills missing values with predefined defaults based on median values derived from the training data. If no features are provided in the POST request, the system uses default vlaues defined in a static dictionary, these include clinical and lifestyle indicators. To test this prediction integration prior to front-end integration, we utilized a sample json to return the result field.

5.2c Evaluation - Front-End Experiments & Results

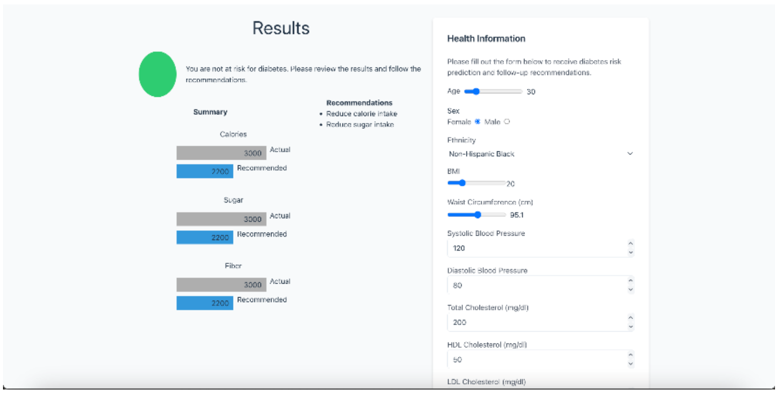


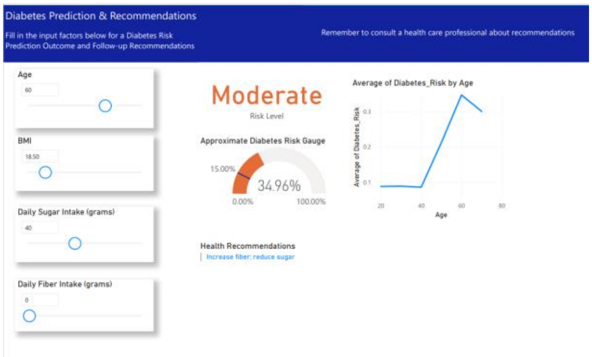
Figure 1: Interactive User Input with Results Page

For ease of demonstration, we elected default values when the user is greated with the input form, i.e., the median values available in the NHANES dataset. This was because the front-end input form is quite extensive and asking for datapoints that users may not have readily available can be tedious. We put additional limitations on the scales of the input variables based on the NHANES data, i.e. the minimum value and maximum value which NHANES allow for each input variable. A key assumption is that the end user would have the necessary laboratory and biometric data available to input, or health care professionals who have access to that data can run the platform for additional clinical decision support.

We ensured the health input form was interactive and intuitive allowing for slider interactions for discrete variables and a combination of radio buttons and dropdown selection for categorical variables. Upon submitting the health input form, this triggers the endpoint of the API to retrieve the diabetes prediction results of XGBoost model. The end user is then greeted with a results page indicating, risk of diabetes, barcharts for comparative results of dietary input such as sugar intake, fibre intake, and caloric consumption alongside text recommendations of what lifestyle changes the user should make. For the results output we kept the visualisations simple and intuitive, providing a stoplight visual for diabetes risk indication. If the user is at risk, an orange stoplight is shown, if the user is not at risk of diabetes, a green stoplight is shown. Currently, this is under the assumption our model provides a binary classification, but we are exploring how to apply probabilistic methodology to XGBoost. If we are successful in applying probabilistic prediction instead of classification prediction, a guage visual may be more suitable for the range of probabilities of diabetes.

As part of the results page we also provide the end user with a comparison to the median values of sugar, fibre, and caloric intake as a way to demonstrate baseline values compared to the individual. These values however are static and in the future we would like to explore a more dynamic visual based on an additional recommendation engine. We opted for barchart visuals as a way for end users to easily interpret where their levels are respective to the baseline values. Although we limited the visual recommendations to sugar, fibre, and caloric intake, in the future we could consider expanding the visuals to include BMI tracking and daily exercise recommendations. What is critical for the comprehension of the visuals are accurate labels, units of measurement, and distinct color indicators. All of which have been applied to our results visualisation page.

We piloted the platform with a sample of <10 users, primarily IT data professionals, to gather feedback on usability and clarity. 67% found the platform easy to understand, suggesting clearer labels on visuals. 83% appreciated the color indicators for health risks but recommended bolding the text for added clarity. While 67% felt confident navigating and inputting data, users proposed clustering questions by type and offering guidance on providing input values. *(See Appendix for further user testing details)*



Due to time constraints we were limited in our development activities thus our visuals in the Vue frontend are not as dynamic as we had intended, however we did also explore visualization using Microsoft Power BI. This is our aspirational end state, where visuals are dynamic, the input for health form is intuitive and not overwhelming, and health risk indicators are clearly marked and consistent with intuition.

Figure 2: Microsoft Power BI Report with Synthetic Prediction Data

6. Conclusions and Discussion

The platform provides clear health risk predictions through intuitive visuals like stoplights, bar charts, and text recommendations. It supports clinical decision-making by enabling health professionals to assess diabetes risks quickly and offers a self-assessment tool for individuals. A responsive and dynamic app could extend these benefits to underserved communities. Challenges with imbalanced NHANES data suggest future efforts to supplement with other sources like the ADA or CDC. The prediction model's reliance on extensive input variables could be streamlined for broader usability. Expanding beyond static classification, a dynamic recommendation engine could enhance personalization based on FDA guidelines. For scalability, transitioning to cloud infrastructure, ensuring compliance with HIPAA and GDPR, and designing mobile-first interfaces are essential steps. Security measures, user feedback integration, and visual enhancements will further improve usability and accessibility. Activities were performed by all team members following similar framework proposed by Stumbo (2013), data acquisition and processing, algorithm development, user interface design, system integration, and testing. As a team, we plan to continue this project further beyond the course to ensure our prototype is in a steady state which can be demonstrated to health care professionals and make disease prevention accessible to those who would benefit from it the most.

Works Cited

1. Ahmed, S., & Ali, H. (2025). Machine learning model for diet recommendation awareness: A clustering and classification approach. *Journal of Healthcare Engineering*, 2025, 5567890.
2. Amiri, P., Tehrani, F. R., & Azizi, F. (2023). Artificial intelligence in personalized nutrition: Promises and pitfalls. Nutrients, 15, 1231. https://doi.org/10.3390/nu15051231

3. Backonja, U., Haynes, S. C., & Kim, K. K. (2018). Data visualizations to support health practitioners' provision of personalized cancer care for individuals with multiple chronic conditions. JMIR Human Factors, 5(4), e11045. https://doi.org/10.2196/11045
4. Brown, L., & Davis, K. (2024). Applications of AI in nutrition: A comprehensive review. *Nutrients*, 16(1), 190. https://doi.org/10.3390/nu16010190
5. Celis-Morales, C., Lara, J., & Mathers, J. C. (2015). Personalising nutritional guidance using 'big data': Current developments and future perspectives. Proceedings of the Nutrition Society, 74(3), 246-255. https://doi.org/10.1017/S0029665114001633
6. Chen, J., Lieffers, J., Bauman, A., Hanning, R., & Allman-Farinelli, M. (2017). The use of smartphone health apps and other mobile health (mHealth) technologies in dietetic practice: A three country study. Journal of Human Nutrition and Dietetics, 30, 439-452. https://doi.org/10.1111/jhn.12446
7. Fulgoni, V. L., Keast, D. R., Bailey, R. L., & Dwyer, J. (2019). Foods, fortificants, and supplements: Where do Americans get their nutrients? The Journal of Nutrition, 141, 1847-1854. https://doi.org/10.3945/jn.111.142257
8. Guo, J., He, Q., & Li, Y. (2024). Machine learning-based prediction of vitamin D deficiency: NHANES 2001-2018. Frontiers in Endocrinology (Lausanne), 15, 1327058–1327058. https://doi.org/10.3389/fendo.2024.1327058
9. Liu, Z., Liu, L., Heidel, R. E., & Zhao, X. (2024). *Exploring nutritional impact on Alzheimer's mortality: An explainable AI approach*. arXiv. https://doi.org/10.48550/arXiv.2405.17502
10. Mattila, E., Orsama, A. L., Ahtinen, A., Vaha-Ypya, H., Manttari, A., Koskinen, E., & Lappalainen, R. (2010). Personal health technologies in employee health promotion: Usage activity, usefulness, and health benefits. International Journal of Medical Informatics, 79(11), 803-813. https://doi.org/10.1016/j.ijmedinf.2010.09.007
11. Mittelstadt, B. D., Allo, P., Taddeo, M., Wachter, S., & Floridi, L. (2016). The ethics of algorithms: Mapping the debate. Big Data & Society, 3, 1-21. https://doi.org/10.1177/2053951716679679
12. Ronto, R., Wu, J. H., Singh, G. M., & Vanderlee, L. (2017). Nutritional quality of fast food items marketed at a global fast food chain and comparison with nutritional guidelines. Nutrition & Dietetics, 74(1), 19-26. https://doi.org/10.1111/1747-0080.12284
13. Schoeppe, S., Alley, S., Van Lippevelde, W., Bray, N. A., Williams, S. L., Duncan, M. J., & Vandelanotte, C. (2017). Smartphone use for health purposes: Systematic review. American Journal of Preventive Medicine, 54, 646-653. https://doi.org/10.1016/j.amepre.2017.01.011
14. Spring, B., Duncan, J. M., Janke, E. A., Kozak, A. T., McFadden, H. G., DeMott, A., ... & Siddique, J. (2013). Integrating technology into standard weight loss treatment: A randomized controlled trial. JAMA Internal Medicine, 173, 105-111. https://doi.org/10.1001/jamainternmed.2013.1221
15. Stumbo, P. J. (2013). New technology in dietary assessment: A review of digital methods in improving food record accuracy. Proceedings of the Nutrition Society, 72, 70-76. https://doi.org/10.1017/S0029665112002911
16. Zhu, Y., Hollis, J. H., & Henry, C. J. (2018). The impact of visualizing eating behaviors on promoting healthy dietary changes: A systematic review and meta-analysis. Nutrients, 10, 799. https://doi.org/10.3390/nu10070799
17. B. Ross Katz, Abdul Khan, James York-Winegar, Alexander J. Titus (2024) NHANES-GCP: Leveraging the Google Cloud Platform and BigQuery ML for reproducible machine learning with data from the National Health and Nutrition Examination Survey https://arxiv.org/abs/2401.06967
18. Yicheng Qi, Ang Li (2024) Causality in the Can: Diet Coke's Impact on Fatness https://arxiv.org/abs/2405.10746
19. Marcos Matabuena, Juan C. Vidal, Rahul Ghosal, Jukka-Pekka Onnela (2024) Deep Learning Framework with Uncertainty Quantification for Survey Data: Assessing and Predicting Diabetes Mellitus Risk in the American Population https://arxiv.org/abs/2403.19752
20. Zhang, Y., & Kaufman, D. R. (2020). Data visualization for health and risk communication. In The Routledge Handbook of Health Communication (3rd ed.). Routledge.

21. Zheyuan Zhang, Yiyang Li, Nhi Ha Lan Le, Zehong Wang, Tianyi Ma, Vincent Galassi, Keerthiram Murugesan, Nuno Moniz, Werner Geyer, Nitesh V Chawla, Chuxu Zhang, Yanfang Ye (2024) NGQA: A Nutritional Graph Question Answering Benchmark for Personalized Health-aware Nutritional Reasoning https://arxiv.org/abs/2412.15547
22. Ziming Liu, Longjian Liu, Robert E. Heidel, Xiaopeng Zhao (2024) Exploring Nutritional Impact on Alzheimer's Mortality: An Explainable AI Approach https://arxiv.org/abs/2405.17502
23. John Semerdjian, Spencer Frank (2017) An Ensemble Classifier for Predicting the Onset of Type II Diabetes https://arxiv.org/abs/1708.07480
24. Centers for Disease Control and Prevention. (n.d.). NHANES data visualization. National Center for Health Statistics. https://www.cdc.gov/nchs/nhanes/visualization/index.htm

Appendix: Additional Visuals:

Statistical Measure	Classification
Precision: 0.0	Borderline Diabetic
F-1 Score: 0.0	Borderline Diabetic
Recall: 0.0	Borderline Diabetic
Precision: .81	Diabetic
F-1 Score: .67	Diabetic
Recall: .57	Diabetic
Precision: .89	Non-Diabetic
F-1 Score: .93	Non-Diabetic
Recall: .98	Non-Diabetic

Table 2. Accuracy measures for Random Forrest Classification

Appendix User Testing Feedback Survey Details:

Survey Details	User 1 Feedback	User 2 Feedback	User 3 Feedback	User 4 Feedback	User 5 Feedback	User 6 Feedback
How easy was it to understand the information presented on the platform?	Very easy	Very easy	Somewhat easy	Somewhat easy	Somewhat easy	Somewhat easy
Optional Comment		May be difficult to access the medical record if I do not have access to it	Additional context for the bar chart visuals for label indicators	Include explicit labels on the visuals to help decipher the output	As a user it is more intuitive to have the input form on the left hand of the page, with the results to follow.	Metric options next to health info (depending on where the user is based) (infographs would be helpful)
Did the color indicators clearly communicate different health risk levels?	Yes	Yes	Yes	Yes	Somewhat	Yes
Optional Comment1			The colors intuitively make sense when it comes to		Include bold text of 'at risk' 'not at risk', 'your result	Red would indicate emotional distress, provide empathetic recommendations

			the disease prediction.		here'. Perhaps a % is more intuitive.	
Was it easy to input your health data into the platform	Somewhat	Yes	Yes	Somewhat	Yes	Yes
Optional Comment2	It would be good to reduce the number of input variables if needed			Need to have the data points on hand as an end user	Long form to input, suggest to cluster the questions, section by section	
Did you feel confident navigating the platform on your own?	Yes	Yes	Yes	Somewhat	Yes	Somewhat
Optional Comment3						Descriptions may be helpful for how to input information
How would you rate the overall look and feel of the platform	Good	Good	Excellent	Average	Good	Average
Optional Comment4						Provide instruction of how to fill in the input
Did anything about the design feel overwhelming or unclear?	Yes		No	No	No	No
If yes, please explain	The number of input variables on the platform, and unsure how the results came to be					

What is one thing you liked most about the platform	clean ui / ux look and feel, very responsive	The graphics upon submitting so that I can understand where I am in diabetes risk factor	Modern and intuitive design, sleek input values, results summary is a good output so as a user I can understand what to do moving forward.			
What is one thing you would improve or change?	Additional visual factors and axis labels.	It may be difficult to understand the intake of carbs, fiber, sugar etc. So there would be necessary training for the user to input this input.		Need to explain to the end user the purpose of entering the information into the platform at the beginning, 'how to' and 'why' information	Form and then the results page , there may not be a need to see everything at once	Extend to additional feature recommendations for when prediction is not green (i.e. for diabetes risk)

Appendix: Additional Literature Reviews which may be applicable further in our project

Mummah, S. A., King, A. C., Gardner, C. D., & Sutton, S. (2016). Iterative development of Vegethon: A theory-based mobile app intervention to increase vegetable consumption. International Journal of Behavioral Nutrition and Physical Activity, 13, 90. https://doi.org/10.1186/s12966-016-0400-z

Dwyer, J. T., Picciano, M. F., & Raiten, D. J. (2003). Future directions for the integrated CSFII-NHANES: What we eat in America-NHANES. The Journal of Nutrition, 133, 576S-581S. https://doi.org/10.1093/jn/133.2.576S

Corey, K. C., Salvatore, A. L., Anderson, E. S., & Haughton, B. L. (2018). Integrating dietary assessment methods in personalized nutrition counseling: A systematic review. Journal of the Academy of Nutrition and Dietetics, 118(9), 1644-1661. https://doi.org/10.1016/j.jand.2018.05.011

Ahluwalia, N., Dwyer, J., Terry, A., Moshfegh, A., & Johnson, C. (2016). Update on NHANES dietary data: Focus on collection, release, analytical considerations, and uses to inform public policy. Advances in Nutrition, 7(1), 121-134. https://doi.org/10.3945/an.115.009258